

Episode Length Is a Hidden Design Choice: Demonstration-Curation Metrics Invert Once You Control For It

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Abstract

Trajectory-smoothness scores are increasingly used to curate imitation-learning demonstrations, on the premise that smoother demonstrations are better ones. We show that on real robomimic data these scores are largely measuring *how long the episode took*, and that the direction of their empirical claim flips once episode duration is held fixed. On can/mh, Spectral Arc Length separates better from worse teleoperators with AUROC 0.952, but episode length alone achieves 0.957, and inside a duration-matched band SAL falls to 0.628 while raw mean jerk, the crude baseline such metrics are motivated against, rises to 0.896. The pattern replicates without retuning on square/mh (SAL 0.898→0.480, jerk 0.431→0.896, length control 0.936→0.407). On a fixed-horizon split where duration cannot carry information, both published metrics sit at or below chance against task success, and curating training data by the contact-aware one costs a third of downstream policy success relative to random selection of the same size (10 seeds, $p=0.0001$ after correction for multiple comparisons). Under a diffusion policy that ordering holds and jerk inverts again, beating both baselines. Whether a duration control is applied is not a robustness detail: it decides the sign of the reported result. We argue it belongs in the minimum reporting set for any demonstration-scoring claim.

1 The design choice underneath the claim

A demonstration-curation metric is evaluated by showing that it ranks known good demonstrations above known bad ones. That evaluation contains a design choice that is almost never reported: whether the good and bad demonstrations differ in *duration*. When they do, and in human teleoperation data they systematically do, because worse operators take longer, any statistic that grows with trajectory length inherits an apparent discriminative power that has nothing to do with what the metric was designed to measure.

This is not a hypothetical failure mode. Prior audit work on a synthetic-defect testbed found that most curation metrics examined achieved near-perfect detection scores by proxying episode length [3], and controlled for it by truncating all trajectories to a fixed prefix. We ask what happens to *named, published* metrics on *real* operator labels when the same control is applied, and find that the control does not merely shrink the effect. It reverses the ranking. The metric that looks best without the control is near chance with it; the crude baseline it was introduced to supersede becomes the strongest detector.

We report three things. (1) On two robomimic tasks, smoothness-metric performance tracks the episode-length baseline almost exactly, and collapses under duration matching. (2) On a fixed-horizon split where duration is constant by construction, both metrics of a recent published method are at or below chance against task success, with the contact-aware metric inverted. (3) Filtering training data by these scores produces policies worse than no filtering at all. The substantive contribution is not that these particular metrics are weak; it is that a single unreported preprocessing decision determines the sign of the conclusion, which makes it a reporting requirement rather than a robustness check.

Related work. RINSE [1] and PSD [5] are recent smoothness- and spectral-power-based curation methods; both validate on success-filtered pools, and neither evaluates on data containing failures. The audit line we build on [2–4] established the detection-vs-curation decoupling and documented the length confound, but on generic metric implementations, synthetically injected defects, and with length controlled by truncation. We extend it to named published metrics, real operator and success labels, and a fixed-horizon split where the confound is excluded by construction rather than by discarding signal. Separately, [7, 8] argue that success rate under-describes policy quality and [9] shows benchmarks admit shortcut solutions; our result is the data-side complement, where the scores used to *select* training data are themselves shortcut-driven.

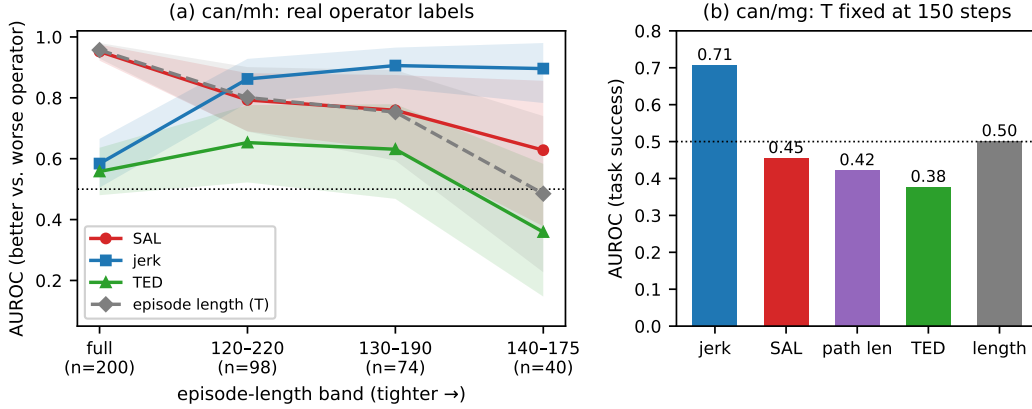


Figure 1: **Smoothness tracks duration, not quality.** (a) Skill AUROC on can/mh across tightening duration bands. SAL (red) follows the episode-length baseline (gray, dashed) down; jerk (blue) rises as the confound is removed; the two cross. In the tightest band length itself is at chance (0.485), confirming the control. (b) On the fixed-horizon can/mg split, duration carries no information by construction; against task success, jerk retains signal while SAL is at chance and TED is inverted.

2 Setup

Metrics. Spectral Arc Length (SAL) and Trajectory-Envelope Distance (TED) as specified by RINSE [1], a recent smoothness-based curation method; TED is reimplemented from its appendix, including the gripper-contact partition that is its distinguishing feature. Baselines: mean jerk (third-difference magnitude of end-effector position) and end-effector path length. All are computed from proprioception at 20 Hz and returned as badness scores. We also report the per-timestep normalization SAL/T , since SAL’s arc length sums over $T/2$ spectral bins and therefore grows with duration by construction.

Data. Public robomimic [6] low-dim. *can/mh* and *square/mh*: 300 successful human demonstrations each with released operator skill tiers (better/okay/worse, 100 each); durations 98–1050 and 123–1051 steps. *can/mg*: 3900 scripted demonstrations, 18.4% successful, every episode exactly 150 steps. Can is the only robomimic task shipping failure-containing splits, so the success-label analysis is confined to it by benchmark availability rather than by choice.

Controls. Two, applied independently. *Normalization*: divide by $T-1$. *Duration matching*: restrict to episode-length bands of increasing tightness, and report the AUROC of episode length itself in each band: a band is only a control where that number is near 0.5. All AUROCs carry 95% bootstrap CIs (2000 resamples); full tables are in the supplementary material.

3 The confound, and what controlling for it does

Uncontrolled, SAL looks excellent, and so does a stopwatch. On can/mh, SAL separates better from worse operators at AUROC 0.952 [0.922, 0.975]. Episode length alone achieves 0.957 [0.929, 0.980]. SAL’s correlation with T is +0.97, and its tier means (100.4 / 57.5 / 44.9) track the tier durations (304 / 181 / 143 steps) almost linearly.

Both controls remove it. Normalization drops SAL to 0.582 [0.497, 0.662] and its length correlation to +0.07. Duration matching does the same thing more directly: across four bands SAL falls monotonically while jerk rises monotonically, and they cross (Table 1, Figure 1a). In the band where episode length is itself at chance, SAL’s CI includes 0.5 and jerk reaches 0.896.

It replicates on a second task, unchanged. Applying the identical pipeline to square/mh, with no retuning: SAL’s duration correlation is +0.96, its uncontrolled AUROC 0.898 against a length baseline of 0.936, and in the controlled band SAL sits at 0.480 while jerk reaches 0.896, the same value it reaches in Can’s tightest band. Two task-specific differences are worth stating rather than smoothing over: jerk is *inverted* on Square before any control (0.431, versus 0.584 on Can), and path length stays strong in Square’s controlled band where it collapsed on Can. The uncontrolled ranking of metrics is task-dependent; the fact that SAL’s advantage does not survive the control is not.

		duration band, loosest $\rightarrow$ tightest			
		full	band 2	band 3	controlled
can/mh	SAL	0.952	0.793	0.759	0.628
	jerk	0.584	0.862	0.906	0.896
	TED	0.558	0.653	0.631	0.359
	length T	0.957	0.801	0.753	0.485
square/mh	SAL	0.898	0.759	0.529	0.480
	jerk	0.431	0.570	0.782	0.896
	TED	0.663	0.685	0.599	0.557
	length T	0.936	0.824	0.613	0.407

Table 1: Better-vs-worse operator AUROC as the duration band tightens, on two tasks. The *length* row is the control: only the rightmost column is duration-matched. SAL falls and jerk rises in lockstep with it, on both tasks. Bootstrap CIs, exact band edges, and sample sizes are in the supplementary material; the tightest bands are small ($n=40$ and $n=30$) and the claim rests on the monotone trend rather than any single cell.

With duration removed by construction, the metrics fail outright. The can/mg split fixes every episode at 150 steps, so no metric can exploit duration and the length baseline is exactly 0.500. Against the real task-success label, jerk reaches 0.706, SAL 0.454, path length 0.421, and TED 0.376 (Figure 1b). TED’s inversion means filtering by TED *preferentially retains failed demonstrations*.

4 Does it matter downstream? Yes, in the harmful direction

Detection AUROC is not what practitioners care about, so we filter can/mg by each metric at a 50% keep fraction, hold the training pool size fixed at 1950 demonstrations, and train GMM behavior-cloning policies (10 seeds each, 50 rollouts per evaluation, best checkpoint per seed). Mean best success rates: no curation 0.380, random 50% subset 0.360, SAL 0.332, jerk 0.332, path length 0.310, TED 0.240. An oracle keeping only true successes reaches 0.892, roughly $2.5\times$ the best metric, so the signal is unambiguously present in the pool and the metrics are not finding it.

Testing those gaps properly (difference-of-means bootstrap, 20,000 resamples, plus Welch two-sample tests, Bonferroni-corrected across eight comparisons at $\alpha = 0.006$): **TED** is below random by -0.120 $[-0.166, -0.076]$, $p = 0.0001$, and below no-curation by -0.140 , $p < 0.0001$; both survive correction. Path length is directionally harmful with weaker support ($p = 0.037$, $p = 0.0071$), clearing conventional significance but not correction. SAL and jerk are below both baselines in the mean but not distinguishable from them ($p \geq 0.06$). We report them that way rather than as evidence of harm: an earlier 5-seed pass appeared to show all four metrics underperforming, and that pattern did not survive the seed expansion. Since this paper’s argument is that under-powered evaluation yields confident claims in the wrong direction, we hold our own results to the same standard. Random and no-curation are themselves indistinguishable ($\Delta = -0.020$, CI spanning zero), so a metric’s deficit is attributable to *which* demonstrations it discards, not to pool size. Full grid and per-comparison tests in supplementary.

Across architectures. A diffusion-policy replication (five conditions, five seeds, matched epoch and evaluation budget; supplementary) inverts one metric and leaves the other in place. TED is worst under both (0.240 BC, 0.092 diffusion), which is what one expects if the deficit belongs to the metric; but at five seeds the grid resolves only differences above 0.054, and TED’s deficit there (-0.028 , CI $[-0.052, +0.000]$) falls below it, so the *ordering* replicates while the *harm* stays established under BC only. Jerk reverses sign, beating both baselines by $+0.096$ and $+0.092$ ($p = 0.0007$, 0.0011 , surviving correction) after being inert under BC: the crude baseline has curation value under the right policy class.

Qualitative inspection explains the direction (Figure 2). On this task successful manipulation requires sustained high-deviation motion through the contact phase, while failure looks like smooth, low-amplitude hovering, a robot that lifts and never commits to a placement. A metric rewarding smoothness is measuring commitment with the sign flipped.

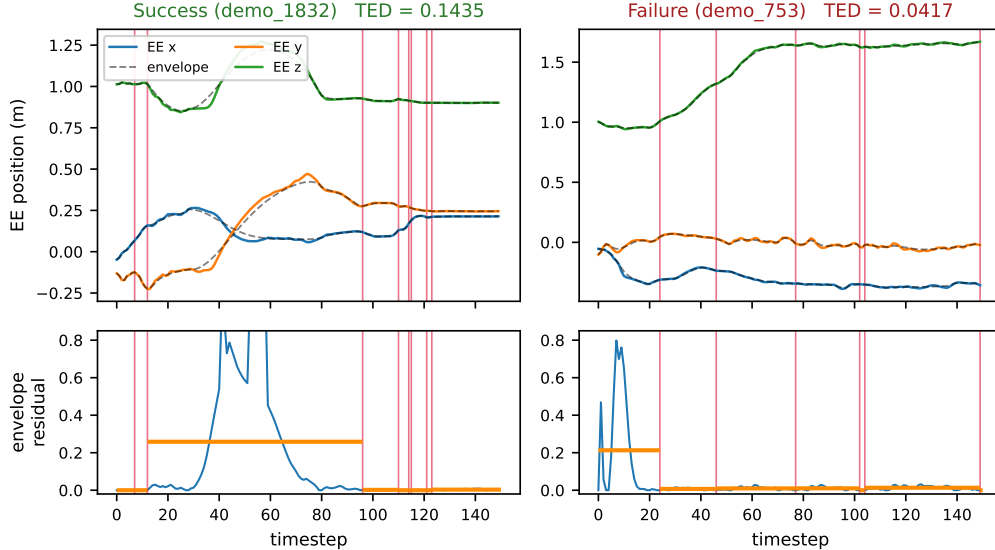


Figure 2: **Failure is smooth; success is not.** A successful (left) and failed (right) can/mg demonstration, with TED’s contact-segment boundaries (red), per-segment Bézier envelope (dashed), and envelope residual (bottom). The success carries one high-deviation segment spanning timesteps 12–96, coinciding with the descend-and-grasp. In the failure the only comparable deviation is the free-space approach (0–24); the remaining 126 steps are a flat tail of repeated gripper cycles that never complete a placement. TED averages over the alignment path and so rates the failure as the cleaner trajectory (0.042 vs. 0.144). Illustrative of the mechanism; the quantitative claim is Figure 1b.

5 What should be reported

The episode-length confound is a substrate effect in a precise sense: it is a property of how the evaluation data was assembled, invisible in the metric’s definition, that determines whether the reported effect points up or down. It is cheap to check and cheap to report, and we suggest three things belong in any demonstration-scoring result.

Report the length baseline. Episode length alone, scored as if it were a curation metric, against the same label. If it matches the proposed metric, the proposed metric has not yet been shown to do anything else. This one number would have flagged the issue in both tasks we examined.

Report a duration-controlled condition, and report its control. A duration-matched band or a normalization, *with* the length AUROC inside that condition, so a reader can verify the control worked. A band where length still scores 0.82 is not a control, and we report our own intermediate bands partly to make that visible.

Separate the label from the deployment target. Skill tiers and task success are different labels and these metrics behave oppositely on them. Curation methods are validated on success-filtered pools and deployed to remove failures from mixed pools; that gap is where the inversion lives.

6 Limitations

Detection results cover two robomimic tasks; the success-label and training results cover one, because Can is the only robomimic task with failure-containing splits, and extending them requires a different benchmark such as LIBERO. Two policy classes downstream (GMM behavior cloning, diffusion); detection results are policy-free. The diffusion grid covers two of the four metrics at five seeds, so TED’s cross-architecture claim is about ordering, not harm, and its absolute rates stay $1.6\text{--}3.1\times$ below BC for reasons we cannot yet identify; the training curves rule out under-training (supplementary). TED is our reimplement from the published appendix, not the authors’ code; it agrees broadly with SAL’s ranking on clean data as a sanity check, but a discrepancy cannot be excluded. The tightest duration bands are small ($n=40$, $n=30$), and we treat them as the endpoint of a trend rather than as standalone evidence. Failures in can/mg are machine-generated; human failure modes may differ, and a human-failure pool is the clearest next step.

References

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Use of AI assistance. An AI assistant was used for code scaffolding, LaTeX drafting, and editing support. All experiments were designed, executed, and verified by the author(s); all reported numbers were produced by the released code and checked against raw outputs.

Supplementary Material

A Full duration-band tables with bootstrap CIs

	full ($n=200$)	$T \in [120, 220]$ ($n=98$)	$T \in [130, 190]$ ($n=74$)	$T \in [140, 175]$ ($n=40$)
SAL	0.952 [.922,.975]	0.793 [.690,.881]	0.759 [.628,.875]	0.628 [.378,.856]
SAL/ T	0.582 [.497,.662]	0.600 [.479,.715]	0.603 [.446,.755]	0.658 [.342,.923]
TED	0.558 [.480,.636]	0.653 [.522,.776]	0.631 [.468,.779]	0.359 [.147,.583]
jerk	0.584 [.505,.665]	0.862 [.783,.928]	0.906 [.832,.965]	0.896 [.783,.980]
path len.	0.783 [.718,.842]	0.590 [.449,.730]	0.624 [.461,.781]	0.429 [.215,.645]
length T	0.957 [.929,.980]	0.801 [.689,.901]	0.753 [.594,.891]	0.485 [.227,.740]

Table 2: can/mh, better-vs-worse operator AUROC, 95% bootstrap CIs (2000 resamples). The tightest band retains 33 better and 7 worse demonstrations.

	full ($n=200$)	$T \in [150, 300]$ ($n=111$)	$T \in [180, 260]$ ($n=55$)	$T \in [200, 240]$ ($n=30$)
SAL	0.898 [.852,.936]	0.759 [.658,.849]	0.529 [.344,.702]	0.480 [.250,.701]
SAL/ T	0.458 [.376,.536]	0.485 [.372,.602]	0.468 [.300,.653]	0.520 [.280,.738]
TED	0.663 [.584,.736]	0.685 [.577,.788]	0.599 [.427,.758]	0.557 [.335,.785]
jerk	0.431 [.349,.509]	0.570 [.455,.687]	0.782 [.618,.923]	0.896 [.724,1.000]
path len.	0.931 [.893,.963]	0.817 [.726,.895]	0.796 [.654,.913]	0.887 [.742,.991]
length T	0.936 [.902,.964]	0.824 [.738,.898]	0.613 [.456,.766]	0.407 [.197,.629]

Table 3: square/mh, same format. The tightest band retains 17 better and 13 worse demonstrations. Note that only the rightmost column has a length AUROC near chance; the two intermediate bands are partial controls and are reported to make the trend visible, not as controls in themselves.

Per-timestep normalizations of TED and path length are omitted from these tables. Unlike SAL, whose raw value grows with the number of spectral bins, these quantities do not scale linearly with T ; dividing by T converts them into inverse-speed proxies that *rise* with skill, which is a different quantity rather than a duration-free measure of smoothness.

B Full downstream training grid

condition	pool size	best success rate (mean $\pm$ std)	95% CI
oracle (successes only)	716	0.892 $\pm$ 0.048	[0.866, 0.922]
no curation	1950	0.380 $\pm$ 0.063	[0.344, 0.420]
random 50%	1950	0.360 $\pm$ 0.061	[0.326, 0.398]
SAL	1950	0.332 $\pm$ 0.061	[0.298, 0.370]
jerk	1950	0.332 $\pm$ 0.041	[0.306, 0.354]
path length	1950	0.310 $\pm$ 0.030	[0.292, 0.328]
TED	1950	0.240 $\pm$ 0.045	[0.212, 0.266]

Table 4: Downstream GMM-BC policy success on can/mg, 10 seeds per condition. Architecture: MLP 1024 $\times$ 1024, 5 GMM modes, batch 100, 600 epochs of 500 gradient steps, learning rate 10^{-4} ; 50 rollouts (horizon 400) every 150 epochs, best checkpoint per seed.

At 10 seeds, differences below roughly 0.05 in mean success rate are not resolvable on this grid. A prior 5-seed version of this analysis reported all four metrics as underperforming both baselines; that pattern held only for TED and path length once seeds were doubled, and we report the corrected result.

Training curves and the choice of estimator. The diffusion runs show no measurable improvement over the second half of training. Averaged across all 25 runs, mean success in the last four of the eight evaluations

metric	vs. random 50%		vs. no curation	
	Δ [95% CI]	p	Δ [95% CI]	p
TED	-0.120 [-0.166, -0.076]	0.0001	-0.140 [-0.186, -0.096]	< 0.0001
path length	-0.050 [-0.090, -0.012]	0.037	-0.070 [-0.112, -0.030]	0.0071
jerk	-0.028 [-0.072, +0.014]	0.25	-0.048 [-0.092, -0.006]	0.060
SAL	-0.028 [-0.080, +0.024]	0.32	-0.048 [-0.098, +0.002]	0.098

Table 5: Difference in mean best success rate against each baseline, with 95% bootstrap CIs on the difference (20,000 resamples) and Welch two-sample p -values. Bold marks comparisons surviving Bonferroni correction across all eight tests ($\alpha = 0.05/8 = 0.006$); only TED clears it. Random vs. no curation: $\Delta = -0.020$, CI $[-0.072, +0.032]$.

condition	GMM-BC (10 seeds)	diffusion (5 seeds)
oracle	0.892 ± 0.048	0.544 ± 0.030
no curation	0.380 ± 0.063	0.120 ± 0.014
random 50%	0.360 ± 0.061	0.116 ± 0.017
jerk	0.332 ± 0.041	0.212 ± 0.030
TED	0.240 ± 0.045	0.092 ± 0.030

Table 6: Mean best success rate for the five conditions run under both policy classes.

exceeds the first four by $+0.017$, against a binomial noise floor of 0.044 for 50-rollout evaluations, and only 3 of 25 runs improve by more than one noise standard deviation; per-run variation across the eight evaluations has a standard deviation $1.01\times$ the noise floor. The same measurement on the BC grid gives $+0.092$, with 56 of 70 runs improving and a ratio of $1.46\times$. The flatness is therefore a property of the diffusion runs rather than of the evaluation protocol, and it rules out under-training as the explanation for the absolute gap. Separately, we report the best checkpoint per seed, a maximum over eight evaluations and so biased upward; the bias is symmetric, since both grids are evaluated eight times. Recomputing with the mean over the eight evaluations leaves all conclusions intact except one worth recording: TED versus no curation under diffusion reaches $p = 0.0110$ and would survive that architecture’s threshold ($\alpha = 0.0125$), where the reported best-of-eight version does not ($p = 0.11$); TED versus random stays non-significant under both. We report the conservative version. Corrections are applied within each architecture; pooling all twelve tests ($\alpha = 0.00417$) changes no conclusion.

C Secondary setting: can/paired

On robomimic’s matched-pair split every metric is inverted against success (TED 0.331, SAL 0.398, jerk 0.368, path length 0.029), but the setting carries an opposing length confound: successful demonstrations are systematically longer (length-alone AUROC 0.194) and path length correlates $+0.86$ with duration. Its matched-pair construction also concentrates the discriminating signal into a handful of terminal timesteps, since each pair shares a near-identical prefix and diverges only at the final placement. Neither property is representative of the mixed-quality pools curation metrics are deployed on, so we report it for completeness and treat can/mg as the primary success-label evidence.

D Qualitative mechanism

Contrasting a successful and a failed can/mg demonstration: both contain one high-deviation segment under TED’s contact partition, but they differ in placement and extent. In the success the high-residual segment spans timesteps 12–96 (84 of 150 steps, mean envelope residual 0.259) and coincides with the descend-and-grasp. In the failure the only comparable deviation is the initial free-space approach (0–24, residual 0.213); the remaining 126 steps are a flat tail in which the end effector holds height while the gripper cycles without completing a placement, every segment scoring below 0.014. Because TED averages deviation over the DTW alignment path, a committed motion spanning half the episode raises the score while a short approach transient is diluted. TED rates the failure as the cleaner trajectory (0.042 vs. 0.144). Per-demonstration TED scores inspected: successes 0.144 and 0.079; failures 0.042, 0.042, 0.017.

metric	vs. random 50%		vs. no curation	
	Δ [95% CI]	p	Δ [95% CI]	p
jerk	+0.096 [+0.068, +0.124]	0.0007	+0.092 [+0.064, +0.116]	0.0011
TED	-0.024 [-0.048, +0.004]	0.17	-0.028 [-0.052, +0.000]	0.11

Table 7: Diffusion-policy grid (Table 6), same procedure as the BC tests above: 95% bootstrap CIs on the difference (20,000 resamples), Welch two-sample p -values, Bonferroni across the four tests reported for this architecture ($\alpha = 0.05/4 = 0.0125$). Random and no curation are indistinguishable ($\Delta = -0.004$, CI $[-0.020, +0.012]$). Policy: conditional U-Net ([256, 512, 1024] down-dims, kernel 5, 8 groups), EMA power 0.75, observation/action/prediction horizons 2/8/16, DDIM with 10 inference steps, batch 256, 600 epochs of 500 gradient steps, lr 10^{-4} , 50 rollouts (horizon 400) every 150 epochs. At five seeds this grid resolves differences of roughly 0.054.

E Reproducibility

All data is public robomimic. Metric implementations, scoring scripts, band analysis, training configurations, and scored CSVs will be released on publication. TED is reimplemented from the published appendix ($w_{\text{ori}} = 0.25$, contact partition at gripper open/close transitions, greedy Bézier envelope per segment, length-normalized DTW distance); SAL follows its published formula. Control rate 20 Hz, $\Delta t = 0.05$ s.